TNPG: Hypergamy project: WeBlog

Super awesome blog hosting and sharing site, where you can update your own blog and see blogs of other users.

SHIP DATE: `2026-11-18 08:00`

by Leon Wong, Marcos Lehman, Wenjie Ma, Efrat Hossain

Program Components

App - `app.py` - Contains all routes and is main entry point to run the application
Authentication logic - `auth.py` - imported into app.py

- Has methods for logging a user in, checking if their account exists, adding a new account to database, logging out.

Viewing blogs of a specific user - `blog_view.py` - imported into app.py

- Has methods to return to the app.py the relevant amount of blog info when viewing someone else's page. This is because if someone has a bajillion blog posts, you don't want to be seeing all of those load at once, so perhaps one page shows the first 10, then the next page shows the next 10, etc.
- Some methods include determining how many posts this person has and which posts to show on what page.

Searching for usernames - `search.py` - imported into app.py

- Facilitates interpreting search queries.
    - If the query perfectly matches a username, go straight to that profile page.
    - Otherwise, show a list of all usernames that contain the query as a substring. (Cap this list at 100 if needed, and provide a button to move on to the next 100.)

`templates/`

- `home.html` - home page.
- `login.html` and `new_user.html` for respective sign in stuff.
- `profile.html` - viewing someone else's blog.
    - Edit and create buttons will always be present, but won't work unless you are the person who owns the blog you are looking at.
    - As a stretch goal, these buttons will only appear if you are the blog owner.
- `edit.html` - edit page for any of your blog posts.

Database Tables

`users`

`(uuid INTEGER PRIMARY KEY, username TEXT, password BLOB)`

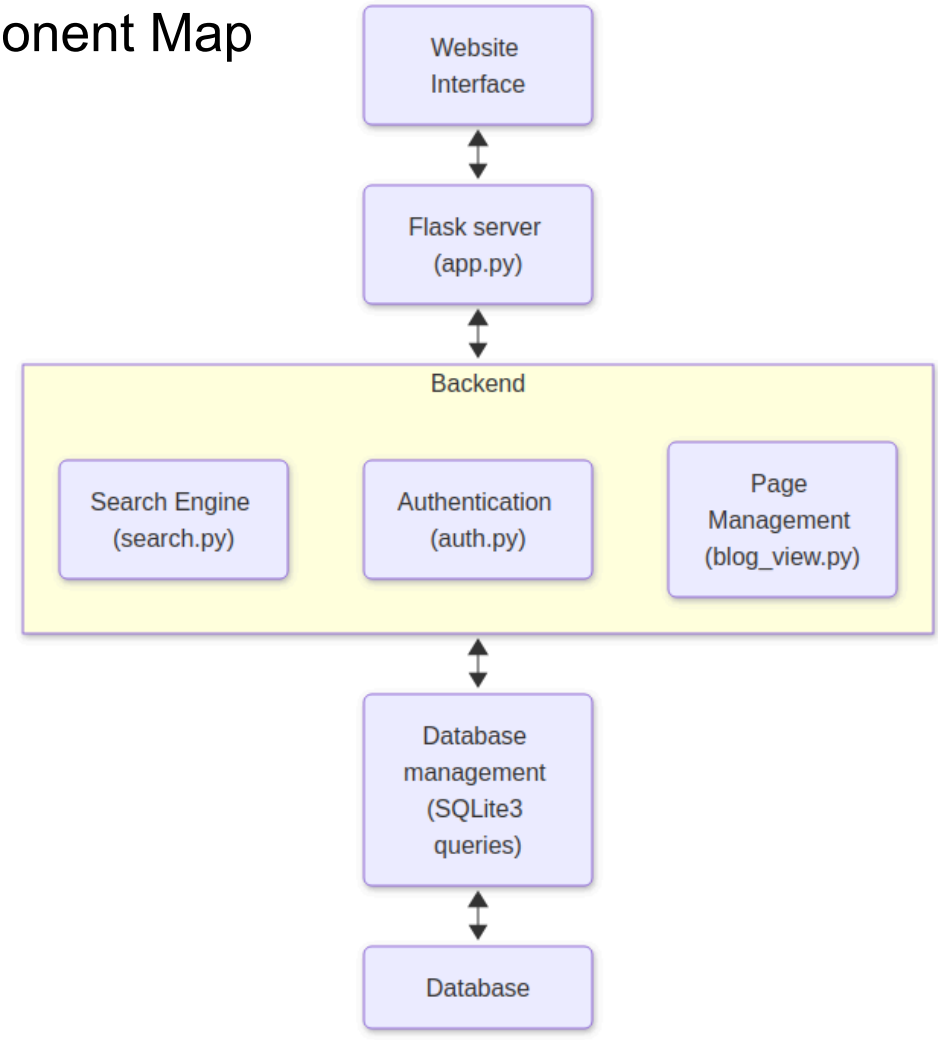
- The password is a blob, because we want to encrypt the password and not affect the resulting bits from the encryption.

`blog_posts`

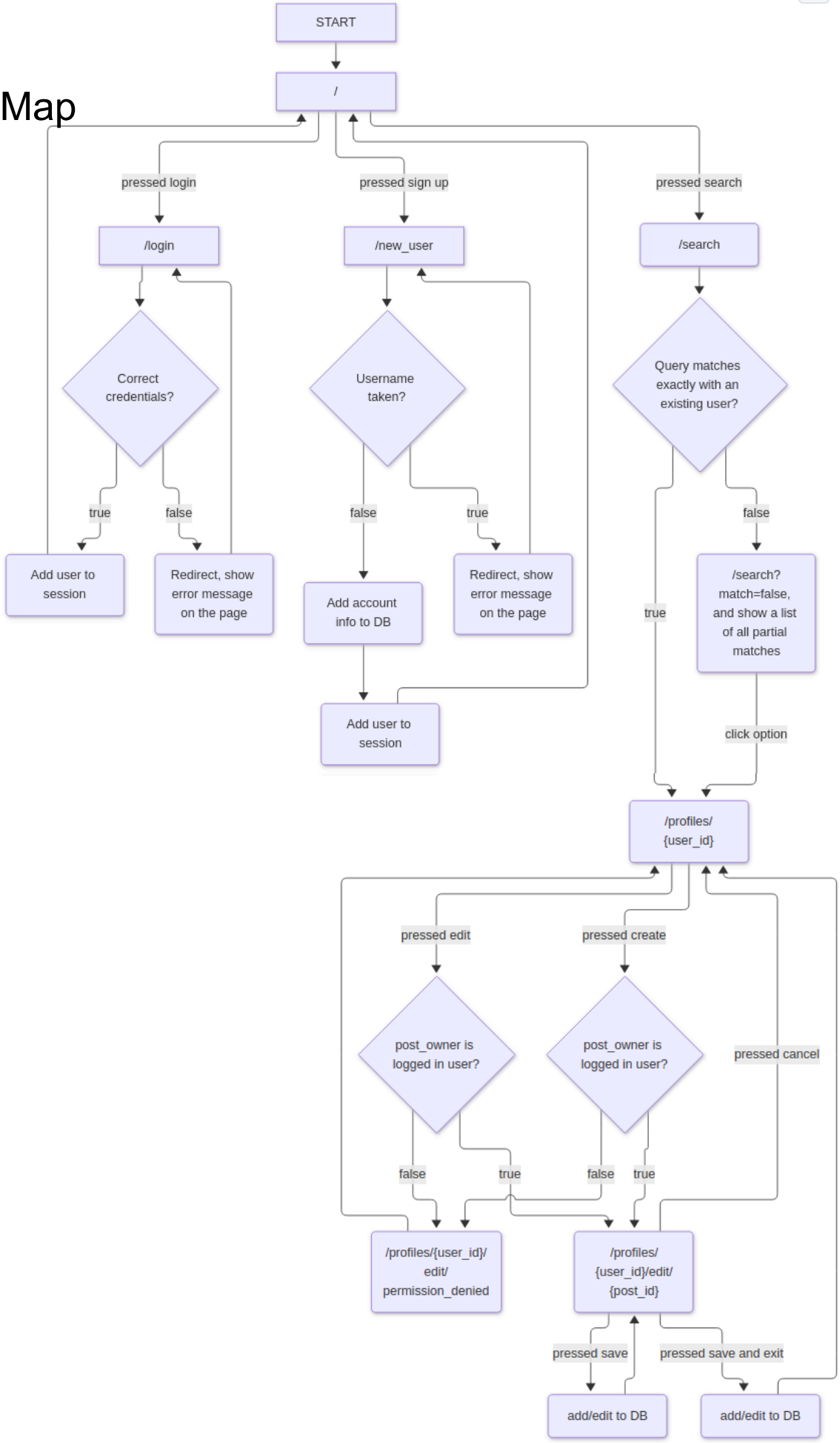
`(pid INTEGER PRIMARY KEY AUTOINCREMENT, owner INTEGER, title TEXT, contents BLOB, published INTEGER)`

- pid = post id
    - It it autoincrement to extra-ensure that the pid's of deleted rows will not be used again.
    - Autoincrement makes it easy to tell the most recent posts. Older posts have lower pids.
- published: 0 = not published (only the creator can see it), 1 = published (anyone can see)

Component Map



Site Map


START
/
pressed login
pressed sign up
pressed search
/login
/new_user
/search
Correct credentials?
Username taken?
Query matches exactly with an existing user?
true
false
false
true
false
true
Add user to session
Redirect, show error message on the page
Add account info to DB
Redirect, show error message on the page
/search? match=false, and show a list of all partial matches
Add user to session
click option
/profiles/ {user_id}
pressed edit
pressed create
post_owner is logged in user?
post_owner is logged in user?
pressed cancel
false
true
false
true
/profiles/{user_id}/ edit/ permission_denied
/profiles/ {user_id}/edit/ {post_id}
pressed save
pressed save and exit
add/edit to DB
add/edit to DB